

Aarya Thakur

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SUMMARY

Final year AI DS student with hands-on experience building LLM-powered systems, agentic AI pipelines, and RAG architectures. Proven in high-pressure environments with 2 internships and 6 hackathon wins.

EDUCATION

Thadomal Shahani Engineering College CGPA 9.43
Bachelor of Engineering in Artificial Intelligence and Data Science November 2022 – June 2026

EXPERIENCE

- Fullstack Intern** June 2025 – July 2025
Augenblick Consulting Remote
- Implemented Deep Zoom Technology, improving large image rendering speed and enabling seamless web-based navigation of gigapixel images.
 - Contributed to building a RAG-powered system that converts RIS files into vector embeddings, enabling accurate research paper classification, tagging, and advanced rule-based filtering.
- AI Intern** January 2025 – April 2025
Nimrobo AI Remote
- Created a tool enabling agentic AI to query and interpret CSV data using SQL for advanced data analysis and insights.
 - Integrated citation tracking into AI Agent outputs, enhancing content credibility and reducing hallucinated outputs.
 - Developed a container orchestration system that automatically spins up isolated AI agent instances with WebSocket access, and automated lifecycle management, improving scalability and reliability.

PROJECTS

- MatryoshkaSearch** [↗](#) | *Retrieval Pipeline with Custom Embeddings* | Python, LangGraph, FastAPI, ChromaDB
- Fine-tuned embedding model on ~7K financial Q&A pairs using Matryoshka Representation Learning achieving 10–13% gains on various metrics over the base model.
 - Built a LangGraph retrieval pipeline with semantic chunking-based document ingestion, query routing and cross-encoder reranking, containerized the full stack with Docker.
- CollabNext** [↗](#) | *Real-Time Collaborative Editor with AI* | Next.js, TypeScript, Firebase, WebSocket, Yjs
- Built a real-time collaborative content studio with WebSockets and Yjs Conflict-free Replicated Data Types, ensuring conflict-free synchronization, and smooth multi-user editing.
 - Integrated AI pipelines for ghostwriting, comic/story generation, and sentiment analysis, streamlining the content creation workflow and accelerating editing pipeline for teams.
- Metaheuristic Job Scheduler** [↗](#) | *Simulated Annealing & Genetic Algorithms* | Python
- Implemented a Simulated Annealing-based solution, enabling escape from local optima and consistently outperforming greedy baseline heuristics.
 - Applied Genetic Algorithm with custom chromosome encoding, crossover, and mutation strategies, achieving faster convergence and superior global exploration than greedy methods.

ACHIEVEMENTS

Winner of **6 Hackathons**, consistently excelling in fast-paced, high-stakes environments across diverse technical domains.

Achieved a **Perfect 10 CGPA** in the 6th Semester.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, SQL
Databases: PostgreSQL, MySQL, Firebase, Vector Databases
Frameworks/Libraries: Next.js, Django, Node.js, Express.js, Tailwind CSS, NumPy, Pandas, PyTorch, LangGraph
Tools/Platforms: Git, GitHub, Docker, Linux, Vercel